Evolving Native-Noise Controllers for Safe Completion with Frozen Manipulation Policies

Completed frozen Jar pilot and separate continuation feasibility screen

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ABSTRACT

Can a compact controller recover safer task completion from a frozen generative manipulation policy under a small interaction budget? Our 176-parameter adapter adds a bounded four-dimensional offset to native noise using measured robot state, executed input history and frozen image/text features. The VLA and native execution contract remain fixed. A 320-call decoder audit on ManiGuard Jar and Clutter passes all 32 same-noise identities and verifies action-prefix influence. Four full development episodes establish integration and timing. A frozen 60-episode study includes matched evolutionary and random-search adaptation and sixteen protected independent checks. All sixty episodes are complete and independently validated. All forty comparison-training trials remain requested-unfinished; four nonviolating lid closures never extend to holding. In the four checking cases per condition, every method has zero safe requested completions. Native and EA each have three violations, versus two for fixed and random search. Their raw goals occur only after violations and fail the requested sequence. Fixed and random finalists have identical retained weights. This pilot establishes no evolutionary safe-completion advantage. A separate 48-call decoder diagnostic finds stronger-offset gripper influence with arm coupling. Four separately frozen engineering trials then test native continuation after a closure-producing controller on an exposed task and seed. All remain requested-unfinished: native and closure-only are nonviolating; both handoffs violate and never satisfy native holding. The privileged trigger fires, without a later ordered milestone. This screen fails its expansion rule. A CPU model-assisted prototype also fails its prediction/calibration audit and remains undeployed. Native-noise and evolutionary VLA steering have precedents. Explained safe completion per total interaction remains a contribution to establish; competent learning comparisons, replication and transfer are still required.

CCS CONCEPTS

• **Computing methodologies**;

KEYWORDS

evolutionary robotics, frozen policies, latent steering, safe manipulation

Research draft, 20 October 2026,
2026.

1 INTRODUCTION

Completing a manipulation task and respecting its physical specification are different outcomes. A robot may transport a jar with its lid open, or avoid every violation while never finishing. ManiGuard makes these distinctions measurable through task checks and passive runtime monitors [5]. We investigate whether a small controller can improve safe requested completion around its family-specific frozen vision-language-action policies.

The adapter steers the generator's native noise before decoding an action chunk, conditioning on current observations, the actual instruction and causal executed action history. Evolution optimizes a small parameter matrix from complete physical simulator outcomes. The VLA, feature encoder and native execution contract remain fixed. This physical comparison deploys no learned safety filter, world model or online action veto. Section 7 separately presents implemented CPU preparation, its failed factual audit and the completed separate continuation screen; those components do not alter the original comparison.

Earlier restraint/feedback pilots found limited trigger opportunities and no safe-completion benefit [3]. Their different controllers and legacy rendering profile are historical motivation, not controls for this experiment.

We separate four questions: does adaptation help native execution; does conditioning help a fixed bias; does evolutionary retention and mutation help relative to other adaptation methods; and do fitted weights transfer? Integration, decoder influence, two adaptation generations and protected checking are complete. The matched Jar comparison followed a protocol frozen before comparative feedback. Its small sample gives a negative exploratory result, without optimizer-superiority or cross-family-transfer evidence.

2 RELATED WORK AND CONTRIBUTION BOUNDARY

DSRL learns in the latent-noise space of a frozen diffusion policy, including generalist-policy adaptation [10]. Flow Reversal Steering maps coarse reference actions through a reversed flow and learns auxiliary noise policies; it also studies repeated or averaged noise across chunk positions [8]. Principal Steering Subspaces constructs a low-dimensional basis from decoder responses and controls its leading directions with SAC [4]. Noise-space adaptation, chunk repetition and low-dimensional steering are therefore not novel by themselves. Our present basis is random and fixed, not response-informed.

Evolutionary VLA steering also has precedents. VLA-Pilot evolves action proposals under an inferred steering objective

during inference [2]. Online-ES uses executed trajectory perturbations and feedback-weighted training to update model parameters [11]. Here the base VLA stays frozen while selection updates a reusable conditional adapter using specification-labelled episode outcomes. This defines an empirical question, not a priority claim. A conventional EA or optimizer substitution alone is insufficient novelty.

The intended contribution is a reproducible, mechanistically explained improvement in safe requested completion per total interaction. Evidence must distinguish useful steering from a tuned constant, and evolutionary advantage from random search and competent ordinary learning. One fitted adapter working with several family checkpoints is the longer-term target. Reusing its interface with per-family refitting is a weaker claim than transferring the same weights.

3 PHYSICAL EVALUATION BOUNDARY

At query time τ, the frozen family policy decodes native Gaussian noise $\epsilon_\tau \in \mathbb{R}^{16\times 32}$ from two RGB views, robot input and the prompt. Native output transforms produce physical joint-target/gripper commands. The first eight are executed unless native termination occurs sooner. The adapter changes noise rather than action horizon. Native is the same family-fine-tuned policy with steering disabled. For Jar, both conditions use the released LoRA checkpoint at step 7400; this pilot performs no new supervised fine-tuning or zero-shot foundation-policy comparison.

Let V be raw native `ltl_violated`, without engagement gating. Requested completion C_{req} requires native success and the pinned family checklist. For Jar, closure must be observed strictly before lifting, remain true at every off-support observation and at the end, and lifting must occur. The final native held/intersection goal must also hold. This stricter sequence check is reported beside the unmodified benchmark verdict. Safe completion and training utility are

$$S_{\text{req}} = C_{\text{req}} \wedge \neg V, \qquad U = \mathbf{1}(S_{\text{req}}) - 2\mathbf{1}(V). \tag{1}$$

A violating completion scores -2; unfinished nonviolating execution scores zero. This is a declared penalized objective. Reporting retains all four joint completion/safety categories, raw violations and denominators. Contact/grasp indicators contextualize unfinished episodes; fewer violations can reflect inactivity.

The monitor supplies evaluation labels and never changes commands. Monitor predicates, object poses, task/family/checkpoint identifiers and goal state are excluded from the adapter. Timeouts are valid outcomes, not infrastructure failures, and do not alone identify their physical cause.

4 NATIVE-NOISE CONTROLLER

4.1 Causal features

The feature vector $\phi_\tau \in [-1,1]^{44}$ comprises a bias, seven arm positions divided by π, two measured finger positions divided by .04 m, seven previous executed environment-input arm targets minus current arm positions divided by .2 rad, and three end-effector displacements since the previous query divided by .05 m. Robot/history coordinates are clipped. These targets precede downstream native-controller clipping; they are neither achieved joint positions nor final motor commands. Initially the previous target equals measured joints and displacement is zero.

The remaining 24 coordinates are eight each from overview RGB, wrist RGB and the actual instruction. Frozen CLIP ViT-B/32 [7] produces 512-dimensional embeddings. Each embedding is normalized to unit Euclidean norm. Fixed Gaussian-QR projections, seeded 20261017, produce eight coordinates, multiplied by $\sqrt{512}$, clipped to $[-5,5]$ and divided by five. No feature fitting uses study outcomes. Images are encoded at every native query and text once per episode, including in the native reference. The conditioning path is implemented; its physical value is unestablished. A recorded-input audit finds only two distinct training instructions and centered text-block rank one across shared initialization. Text connectivity therefore does not establish semantic generalization rather than case-specific offsets.

4.2 Bounded steering with native stochasticity

For genome $W \in [-1,1]^{4\times 44}$, define $c_\tau = \tanh(W\phi_\tau)$ and

$$z_{\tau,h} = \epsilon_{\tau,h} + .35Bc_\tau, \qquad h = 1,\ldots,16. \tag{2}$$

The first seven rows of $B \in \mathbb{R}^{32\times 4}$ are a reduced QR basis of a Gaussian 7×4 matrix with PCG64 seed 20261019. Remaining rows are zero. The offset repeats across the chunk and has Euclidean norm at most $.35\sqrt{4} = .70$ per position in noise coordinates. This bounds sampling input, not joint motion or violation probability. Gripper/padding noise are not directly steered, but decoder coupling can affect their outputs. No persistence, learned noise scale or activation gate is used.

For $W = 0$, an explicit branch calls unchanged native `Policy.infer` without an explicit-noise argument. Nonzero controllers preserve the native RNG split and Gaussian draw before adding the offset. A zero noise tensor would define a different policy. Receipts retain RNG transitions, noise/observation hashes, offsets, decoded physical chunks and source hashes. The runtime logs steered proposals and final environment inputs. It does not buy an additional unsteered chunk at every query; same-noise comparisons are confined to the bounded decoder audit.

4.3 Matched search

Shared initialization evaluates identity, positive and negative constants, and a uniform random conditional matrix on two cases. Constants set only $W_{0,0} = \pm\operatorname{atanh}(.5)$, yielding $\pm.175$ along the first basis direction. The random matrix uses seed 20261019. Both methods receive these same eight episodes, counted once globally and attributed to both information budgets. The better constant becomes the fixed reference, with ties selecting positive. This tunes two declared constants, not an extensive fixed-policy baseline.

Each method evaluates two generations of four candidates on two common cases per generation. Both freshly re-evaluate the previous cohort's best candidate. EA adds three children from alternating top-two parents using iid Gaussian weight mutation with standard deviation .25 and clipping to $[-1,1]$. Random search instead draws three uniform matrices from the same domain, with the same retained-incumbent cost. Ranking uses summed utility and stable identifiers for ties. Only the last complete cohort supplies each finalist, preventing best-of-repeat promotion. Both

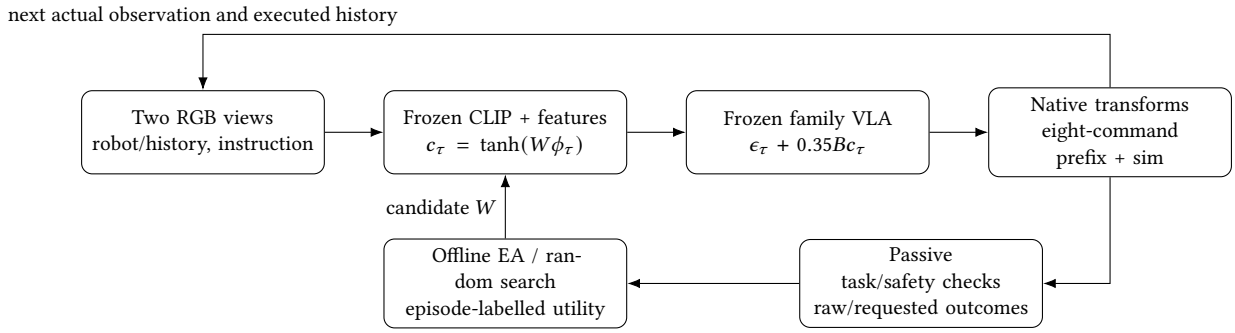


Figure 1: Implemented information boundary. Candidate weights are fixed within an episode; the native Gaussian draw remains present. Outcome labels guide selection of later candidates without vetoing execution. All conditions pay the same feature extraction cost. The VLA also receives its unchanged native observation inputs. No learned safety filter or world model is deployed.

Algorithm 1 Frozen comparison after four development episodes

Require: Pinned policy, features, domain, cases, seeds and budget
Evaluate four shared controllers on two cases (8 episodes)
Select fixed reference from the two constants
Initialize EA and random search with the same scored data
for generation $g = 1, 2$ **do**
    Propose both four-candidate cohorts, including incumbents
    for candidate slot $j = 0, 1, 2, 3$ **do**
        for method in the frozen alternating slot order **do**
            Evaluate its candidate on two common cases
            Verify both physical receipts and record scores
        end for
    end for
    Rank each complete cohort for subsequent selection
    Preserve all attempts, ancestry, RNG and physical records
end for
Seal native, fixed and the two final-cohort finalists
Check four conditions on four blocks (16 episodes)
Report outcomes, block differences, activity and costs

method RNGs use seed 20261020. Descendant execution interleaves methods at each candidate slot, using EA/random at even slots and random/EA at odd slots. Lineage and pending/completed evaluations are persisted.

5 PROSPECTIVELY FROZEN PHYSICAL PROTOCOL

5.1 Scope and native execution

Jar and Clutter provide development exposure; Cabinet and Dusty remain physically untouched by this study. Jar/Stack/Lid/Clutter have programme-level history. Jar is the primary pilot and Clutter the declared capacity fallback. The selected protocol was frozen at 04:27:23 UTC on 20 October before comparative feedback, using development timing and the authority criterion. It selects Jar. This pilot does not train or test one shared cross-family adapter.

Execution uses native 640 × 640 DLAA rendering, two actual camera views, the verified extended-finger Franka, assisted grasping, 20 Hz policy control, 60 Hz rendering and 120 Hz physics. Hard horizons are 2,000 controls for Jar and 1,100 for Clutter. Preprocessing, gripper binarization, native clipping and termination are unchanged. Jar uses checkpoint step 7400, revision `1d84eda07031`; Clutter uses step 7100, `f572e33bdb49`. Full revisions and file hashes remain in study records.

Jar training tasks are 0019/0009, with base seeds $6101910 + 10g + i$, generation $g = 0, 1, 2$ and task index $i = 0, 1$. Checking tasks 0016/0012/0011/0003 use seeds $6101940 + i$. These blocks are reserved from this optimizer's feedback. Historical validation allocation does not establish absence of earlier programme exposure; we conservatively describe them as exposed tasks with independent checking. The original main-study test partition remains excluded. Native per-scene sampling-seed derivation is retained.

5.2 Allocation and hardware balancing

Table 1 specifies the 60 planned valid full episodes. At most four genuine infrastructure failures may be replaced, yielding a 64-full-attempt ceiling. Every partial/invalid attempt and discarded candidate is charged. Valid timeouts and adverse scores cannot be replaced. Separate engineering used six starts: three failed during preparation and three reached 32 controls each. The 96-control engineering allowance is exhausted. Physics before callback attachment is unknown, not zero. The original 320-call decoder allowance is exhausted. A separately authorized 48-call follow-up, reported below, changes neither this physical allocation nor the controller used in the frozen comparison.

Two workers each colocate simulation and policy on one RTX 3090. Training case i uses worker $(i + g)$ mod 2 for both methods. Denoting native, fixed, EA and random by N, F, E, R, checking orders are (N, F, E, R), (F, R, N, E), (E, N, R, F) and (R, E, F, N). Adjacent pairs run concurrently; positions zero and two use worker zero. Every condition therefore runs twice per GPU. One admission ledger enforces the global attempt ceiling. Process identities and terminal receipts permit reattachment without duplicate physical launches.

Table 1: Frozen full-episode allocation. Shared initialization is acquired once. The sixteen-episode independent check is protected from adaptation and tuning.

Stage	Valid episodes
Development, Jar + Clutter	4
Shared initialization: four controllers, two cases	8
EA: two generations, four candidates, two cases	16
Random search: same structure and domain	16
Independent check: four conditions, four blocks	16
Planned total	60
Maximum starts with infrastructure replacements	64

Capacity protects 56 post-development episodes plus four replacement slots as 30 two-worker waves. Its per-wave bound is the maximum of 180 s, 1.25 times the slowest complete development process, and a startup-plus-native-horizon throughput estimate with the same margin. Including 900 s for lifecycle/preservation, Jar requires 15.008 h. The original window was insufficient; the user subsequently authorized the full comparison on this instance. The fixed cutoff is 19:30 UTC and independent local stop 20:00 UTC on 20 October. Horizons/rendering were not reduced to fit.

5.3 Analysis and uncertainty

The final comparison retains all four requested completion and safety categories and raw benchmark completion per condition, with paired differences by checking block. Initial robot/image hashes are checked: common seeds do not prove identical camera observations. Contact/grasp indicators contextualize noncompletion. Cost curves use common cumulative physical control counts and retain all adaptation, failures and discarded candidates. Model calls, encoder work, elapsed time and allocated GPU-hours are separate counters. Shared episodes count once physically and once within each method's information budget.

Setup, decoder calls, idle allocation and preservation are charged. GPU-hour and running-cost estimates are upper bounds over elapsed wall time from the original provider start, counting both GPUs. The own-instance API does not expose verified billed runtime; a container restart is not credited as unbilled time. Four checking episodes per condition and one optimizer seed provide descriptive effect sizes, not a superiority or safety guarantee. Zero violations in n independent episodes implies an approximate 95% upper rate bound of $3/n$, not zero; four episodes are particularly weak evidence.

6 COMPLETED PHYSICAL AND DECODER EVIDENCE

The complete original60 export is dated 20 October, after all forty training, four development and sixteen protected checking trials completed. Its full design verification and private-archive physical replay pass. The separately frozen four-start engineering screen is also complete and is reported separately below.

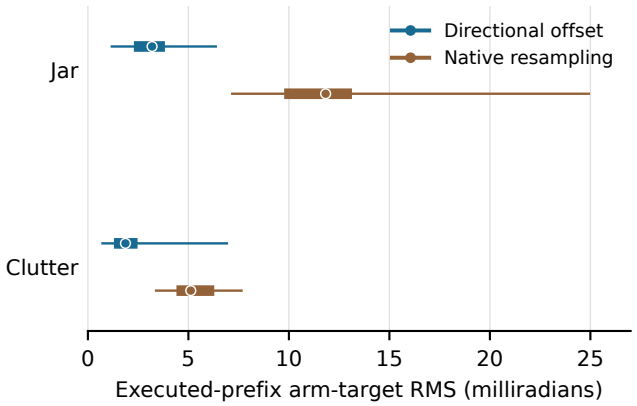


Figure 2: Arm-target RMS differences over the executed eight-command prefix. Thick intervals show quartiles, thin intervals the range and circles the median. There are 128 directional comparisons and eight native-resampling comparisons per family, clustered within eight saved observations. These are decoder responses, not independent physical trials or safety measurements.

6.1 Identity and decoder influence

Eight saved observations per family, from two development tasks, each use two native noise draws. Each draw receives two same-noise baselines and positive/negative offsets along four directions: ten calls and 320 total. All sixteen identity pairs per family pass the declared 10^{-5} absolute/relative tolerance. Full remote recovery also reproduces noise hashes and RNG transitions without new model calls.

Perturbed-prefix arm RMS ranges from .001131 to .006422 rad for Jar, median .003193, and .000666 to .006975 rad for Clutter, median .001868 (Figure 2). Native-resampling medians are .011827/.005118 rad. All four directions per family pass the frozen criterion: observation-level mean RMS at least .001 rad on at least half the observations, with two passing directions required. The smallest individual Clutter response need not exceed this observation-level threshold. No executed-prefix gripper flips occur among 128 directional probes per family. Coupling elsewhere is not excluded.

A 32-control Clutter integration check uses an untrained dense conditional matrix, exercising all 176 weights including 96 semantic weights. Removing semantic features offline changes its code by RMS .3781 on factual observations. This verifies a software path, not semantic competence or a physical counterfactual.

6.2 Full development outcomes

Table 2 preserves four full episodes on different tasks/seeds. Their native/bias differences do not estimate an adaptation effect. Jar native reaches the raw benchmark goal but fails the closed-lid sequence: requested unsafe unfinished, versus raw unsafe completion. Jar bias and Clutter native reach their full horizons, both valid outcomes rather than replaceable failures.

Development closeout contains ten starts, 4,394 controls, 4,464 environment steps and 26,987 known physics ticks. There are 551

Table 2: Full development episodes. C is requested completion, V raw violation, A controls and Q policy calls. Each row uses a different task/seed.

Family/controller	Task	A	Q	C	V	U
Jar native	0013	666	84	0	1	−2
Jar positive bias	0017	2000	250	0	0	0
Clutter native	0039	1100	138	0	1	−2
Clutter negative bias	0010	532	67	1	0	1

Table 3: Shared training initialization, two episodes per controller. SC/UC are safe/unsafe requested completion; SU/UU are safe/unsafe unfinished episodes. V counts violations and G episodes with a recorded grasp. Random here is the shared untrained conditional initializer, not a selected random-search finalist.

Controller	SC	UC	SU	UU	V	G
Native	0	0	1	1	1	1
Positive constant	0	0	1	1	1	0
Negative constant	0	0	2	0	0	0
Random conditional	0	0	0	2	2	0

physical-policy calls and 320 decoder calls. Encoding records 1,102 images, seven text calls, 23.519 s extraction and 38.443 s loading; physical-policy request wall time is 293.776 s. These totals precede the comparison and are not final study costs. Callback-unobserved initialization physics remains unknown.

Clutter native has a 574.6 s cold startup versus its partner's 99.9 s. Logs identify a locked shared Kit cache and shader compilation. The episode then completes normally and retains its cost in capacity selection. This technical delay is distinct from its valid adverse physical outcome.

6.3 Completed shared initialization

All eight shared initialization episodes reach the full 2,000-control horizon without requested or raw benchmark completion (Table 3). Each controller receives the same two nominal task/seed cases, costing 4,000 controls and 500 policy calls. Shared acquisition therefore costs 16,000 controls and 2,000 policy calls globally, attributed once within each optimizer's information budget.

The negative constant's utility sum zero leads native and positive at −2 and the random initializer at −4. It never grasps and contacts the target in only one of its two episodes. This lead concerns nonviolating unfinished execution, not improved safe completion. The frozen rule selects it as the fixed reference and initial incumbent of both searches. Native task 0019 goes off support while open at action 537; closure first becomes true at 638. Positive task 0009 instead loses uprightness at 1116 while still supported, with no recorded grasp. These are factual failure descriptions, not counterfactual explanations of a steering effect.

The initial robot input matches native in all six comparisons of a different controller on the same case. Both RGB views differ in every pair. Stored native-noise hashes and RNG transitions agree at all 1,500 paired query indices across those six streams. The common random stream is realized while observations remain different. These are correlated consistency checks over two cases, not independent outcome trials.

Table 4: Supplemental gripper-noise influence. Entries count saved observations with at least one binarized gripper-command flip in the first eight decoded actions under either sign. The last column is the largest arm-target difference across those positions and the ±2.0 probes. These are correlated decoder inputs, not physical outcome trials.

Family	Magnitude 0.5	Magnitude 2.0	Max arm (rad)
Jar	0/4	3/4	.05875
Clutter	0/4	4/4	.01773

6.4 Exploratory gripper-coordinate diagnostic

Four generation-one incumbent rechecks subsequently evaluate the same negative constant on the new training seed panel. Each method scores $[0, -2]$: no requested or raw completion and one violation per pair. At this diagnostic cutoff, all twelve comparison episodes are unfinished, six violate, and only one ever grasps. Eight never command gripper closure; the other four each show actual finger closure. None of their arm proposals changes at the outer environment bounds. These observations did not identify a unique failure cause. Offspring evaluation was still pending at that earlier diagnostic boundary; its completed first generation is reported below.

The frozen basis excludes direct support on the gripper-noise coordinate. During a diagnostic pause, a separately frozen 48-call follow-up tests coordinate seven (zero-based) on eight saved middle/last development frames, four per family from two episodes. Each input receives a native draw, an explicit same-noise repeat and offsets ±0.5 and ±2.0 on that coordinate at all sixteen chunk positions. No simulator or encoder runs. This raises total decoder requests to 368 while retaining the original audit and physical cohort unchanged. All eight additional same-noise identities pass at the original 10^{-5} tolerance.

The stronger offset changes gripper commands in both families (Table 4), with accompanying arm changes. Its positive sign opens the gripper in all four Clutter frames, including two factual held-target frames; its negative sign closes it in the two unfinished Jar-bias frames. Conditional timing and physical benefit remain untested. Three inputs already lie on violating factual trajectories, and no Jar input demonstrates a successful closure transition. A flip there cannot establish prospective violation prevention. This is influence in a candidate new representation, with no repair applied to the frozen physical controller.

An independent offline sensitivity analysis uses 3,334 factual Jar query rows and 205 Clutter rows from completed development/training episodes. The three persisted EA mutations change their parent codes by RMS .5410/.5825/.4130 on the Jar pool; only .990%/1.110%/.127% of their code components have magnitude at least .95. Inactive mutation or general tanh saturation is therefore unsupported as the current explanation. These are noise-space

Table 5: Eight paid training episodes per row, including incumbent rechecks. Requested safe/unsafe completions are zero throughout. SU/UU are safe/unsafe unfinished; every UU has a raw violation. G counts native target holding, L nonviolating supported closure and B raw benchmark completion. The sole raw goal is unsafe and requested-unfinished.

Gen.	Method	SU	UU	G	L	B
1	EA	4	4	0	1	0
1	Random	4	4	0	1	0
2	EA	6	2	0	1	0
2	Random	6	2	1	1	1

sensitivities on correlated factual inputs, with longer trajectories weighted more. They do not measure decoded actions, semantic competence or physical counterfactual benefit.

6.5 Completed training and ordered progress

All sixteen first-generation episodes reach 2,000 controls and 250 policy calls. Neither method records requested or raw completion or a grasp. Each has four violating and four nonviolating unfinished episodes, seven contact episodes and population utility sum −8 (Table 5).

The last EA child wins the cohort with utility zero, versus random search's best −2. Its first case closes the lid on support at observation 1,187 but never grasps; its second never contacts the jar. This is a training-utility lead through nonviolating unfinished execution, not a safe-completion benefit. Random draw 2 closes safely at observation 482 and remains supported and closed through 2,000, but likewise never grasps. It ties all random candidates under the frozen utility; stable identifier ordering correctly retains the incumbent rather than rewarding this undeclared milestone.

Across the eight shared initial and sixteen first-generation episodes, none completes, twelve violate and only the initial native episode ever grasps. The twelve safe unfinished episodes comprise ten without an ordered milestone and two with supported lid closure only. Nine comparison episodes issue closing gripper commands, but none after safe supported closure. These factual complementary partial behaviours motivate a separately declared continuation test; they do not establish composability or uniquely diagnose failure. Milestone rewards, tie-rule changes and transfer of a random-search controller into EA would change the method and are excluded from this comparison.

On the second frozen seed panel, both methods have two violations, six safe unfinished episodes and population utility −4. EA contacts the jar in six cases and random search in five. Both incumbents score $[0, 0]$ without closure or holding; one case in each never contacts the jar. The earlier EA utility lead does not persist on this common panel. New EA/random children close safely at observations 1,157/1,115, again without holding. All four nonviolating closure examples concern task 0019 across two nominal seed panels, using different matrices.

Random's last child on task 0009 first loses uprightness at 788 while supported, before holding at 833 and lift at 837. Closure never occurs. Raw goal at 1021 and native termination at 1030 follow that violation, so utility remains −2. This demonstrates holding in an unsafe context, not safe composition with another case's closure. Across all forty comparison-training trials, sixteen violate and twenty-four are safe unfinished; none completes the requested sequence. The sealed EA finalist retains its first-generation winner; random retains the negative constant, identical to the fixed reference. Selection uses only the frozen utility and identifier tie rule. All sixteen independent checks are now complete.

Table 6: Independent checking, four episodes per condition. SC/UC and SU/UU retain requested completion/safety categories. V counts violating episodes, B raw benchmark goals and G episodes satisfying the native holding predicate, without an independent body-link grasp witness.

Condition	SC	UC	SU	UU	V	B	G	Mean U
Native	0	0	1	3	3	1	1	−1.5
Fixed	0	0	2	2	2	0	0	−1.0
EA	0	0	1	3	3	1	1	−1.5
Random	0	0	2	2	2	0	0	−1.0

6.6 Completed independent checking

Table 6 reports the sealed finalists on four nominally paired Jar task/seed blocks, tasks 0016, 0012, 0011 and 0003. These cases were protected from this optimizer's training, but retain the programme's earlier exposure; they are not the untouched main-study test set. Every requested completion is false. Native's raw goal on task 0003 and EA's on task 0012 follow violations and remain requested-unfinished.

The six candidate-minus-reference mean utility differences are fixed/native +0.5, EA/native 0, random/native +0.5, EA/fixed −0.5, random/fixed 0 and random/EA +0.5. Every safe-completion difference is zero. Nonzero utility differences arise solely from the violation difference on task 0011. Native and fixed contact the target in all four cases, EA and random in three. This does not establish why the differing failures occur. Fixed and random are the same genotype: random retains the initial fixed controller through both generations; EA retains its first-generation child unchanged in generation two. These are retained lineages, not independent discoveries. Four blocks and one optimizer run establish no evolutionary benefit or safety-rate guarantee.

The eight-value initial robot inputs match within every block. All non-native initial camera pairs differ at some pixels, but each camera's mean absolute channel difference, with RGB normalized to $[0, 1]$, is below $1/255$. Ten of twelve case comparisons have both camera 99th-percentile differences at most $1/255$. Saved-view inspection finds no visible object or camera relocation. Full simulator-state equality, the cause of pixel variation and its policy sensitivity remain unestablished.

All sixty valid episodes plus six engineering starts, including failures, cost 112,131 policy controls, 14,019 physical-policy requests, 112,761 returned environment steps and 678,393 known physics ticks. The separate 368 decoder requests bring neural policy returns to 14,387, with no failed or incomplete server return.

Feature extraction records 28,038 images and 63 text encodings. The manual short engineering start is counted once; pre-callback initialization physics remains unknown. At the original accounting boundary, elapsed rental time gives 41.091 allocated GPU-hours over both GPUs and a $12.586 upper-bound estimate at the observed rate, including setup, idle time and preservation. This is not verified billing or final rental closure. Later analysis and the separate follow-up retain their own allocation accounting. CPU work is separate.

7 CONTINUATION SCREEN AND MODEL-ASSISTED PREPARATION

Status: four physical engineering trials complete; CPU model fails qualification. These separate investigations change no original candidate, objective or checking result. The physical screen tests a composition hypothesis on one exposed case; the learned model and action-changing filter remain undeployed.

7.1 Prospectively frozen continuation method

Before original independent checking, we froze random-search draw 2 from generation one, task 0019/base and exposed seed 6101920. Its earlier factual closure at 482 motivates this engineering case. All four references share that selected matrix and development information: native only, closure only, fixed-clock handoff and privileged supported-closure handoff. This tests feasibility, without a fresh-seed generalization or EA claim. The clock adds elapsed-query information beyond the original 44 actor coordinates; privileged gating additionally uses monitor labels.

The privileged trigger requires three consecutive post-control observations with closed lid, support and no preceding violation. Observation zero is excluded. Loss of qualification before the next eight-control query cancels the trigger. Switch there and latch native execution, preserving the policy Gaussian stream, episode identity and seed. The fixed clock is 488 from the exposed example. Native and closure-only references make no switch. Every condition retains the same encoder schedule, native termination and full 2,000-control horizon.

The four-total-start cap includes partial failures, with no automatic replacement or extra decoder probe. Each start has independent startup/full-wall guardians, durable admission and exclusive policy-service ownership. Original60 completion, full restoration, unchanged source and fresh machine/rate/armed-clock/capacity checks passed before dispatch. The compound reader verifies actual components, RNG continuity, executed commands, exact native monitor entries and captured server returns. Software preparation used 74 synthetic sampler calls, separately from the 368 neural decoder-only calls.

A later stage requires a positive closure prefix and actual native continuation, no whole-episode violation, and a new ordered holding/lift/transport milestone strictly after switching or first safe requested completion after switching. At-switch observations belong to the prefix. Holding uses the native predicate without an independent body-link grasp witness. This criterion authorizes no automatic extra starts; none of the actual references meets it.

7.2 Completed physical screen

All four starts reach 2,000 controls and pass full verification, with no invalid start (Table 7). No requested/raw goal or native holding occurs. Native never contacts the target. Closure-only contacts it and first closes at observation 596, remaining nonviolating with 1,405 closed observations, without holding. This repeats partial closure on the selected exposed seed.

Clock execution loses uprightness while still supported and open at observation 475, before switching at 488. Its later first closure at 531 follows that violation and earns no safe ordered credit. Privileged execution closes at 445, debounces at 447 and switches at 448. It loses uprightness at observation 471, 23 native controls later, while still closed and supported. These timings do not identify an action-specific cause; clock's first violation cannot be attributed to its later handoff. Neither trial adds ordered holding, lift or transport. There is no late native-goal confirmation to satisfy the alternative completion-after-switch criterion. The screen is not expanded.

Native and closure-only command opening throughout, with finger joints near .04 m. The handoffs issue 40/142 closing inputs, first at control 1456/642, respectively, after their first violations. Measured joints later reach approximately zero, and native holding remains absent (Figure 3). The observed failures therefore require more than an account of uniformly opening commands. Closure timing also differs across nominally matched streams. Unequal initial camera images and the single selected case limit causal interpretation; full reset-state identity is unproven. A timeout does not establish irreversibility or a unique perception, dynamics or controller defect.

This screen adds 8,000 controls, 1,000 physical policy requests, 8,040 returned environment steps, 48,356 known physics ticks and 2,000 encoder images. Across both protocols and all engineering attempts, totals are 120,131 controls, 15,019 physical policy requests plus 368 decoder-only requests, 726,749 known physics ticks and 30,038 encoder images. All 15,387 neural-policy returns reconcile exactly; all nine original service prefixes remain unchanged. At the dated 16:57 UTC allocation snapshot, both GPUs account for 43.179 allocated GPU-hours and a $13.226 upper-bound estimate, without verified billing or final rental closure. Later preservation time remains additional.

7.3 CPU model audit and rejected deployment

A five-member episode-bootstrap ridge ensemble predicts the next 44 compact features from 153 inputs: current/previous features, a history-valid flag and eight executed eight-dimensional commands. It uses 32 fixed nonlinear random features and separate heads for a new violation and five task predicates. Actual-future-feature readouts distinguish readout error from imagined-dynamics error. The model generates no future RGB or complete closed-loop VLA episode. Prepared residual/filter, candidate-allocation and ROSARL interfaces are CPU prototypes. ROSARL needs a declared external value estimator and frozen practical penalty [9]; there is no trained ROSARL task-policy baseline here. World-model safety planning already has precedents [1].

The audit was frozen before fitting. Eight shared-initial episodes on tasks 0019/0009 supply 1,992 fitting transitions; development

Table 7: Separate engineering screen: one full trial per mode, on the same exposed task/seed. SC/UC and SU/UU are requested completion/safety categories; V is the whole-episode violation. Stage is the highest ordered milestone before any violation; stage 1 is supported closure. A_{post} counts actual native controls after a switch; native-only has no switch and executes 2,000 native controls. t_V is the first violating observation; N_{close} counts executed closing inputs. These four cases establish no rate, causal effect or transfer.

Mode	SC	UC	SU	UU	V	Stage	Switch	A_{post}	t_V	N_{close}
Native	0	0	1	0	0	0	n/a	0	n/a	0
Closure-only	0	0	1	0	0	1	n/a	0	n/a	0
Clock handoff	0	0	0	1	1	0	488	1512	475	40
Privileged handoff	0	0	0	1	1	1	448	1552	471	142

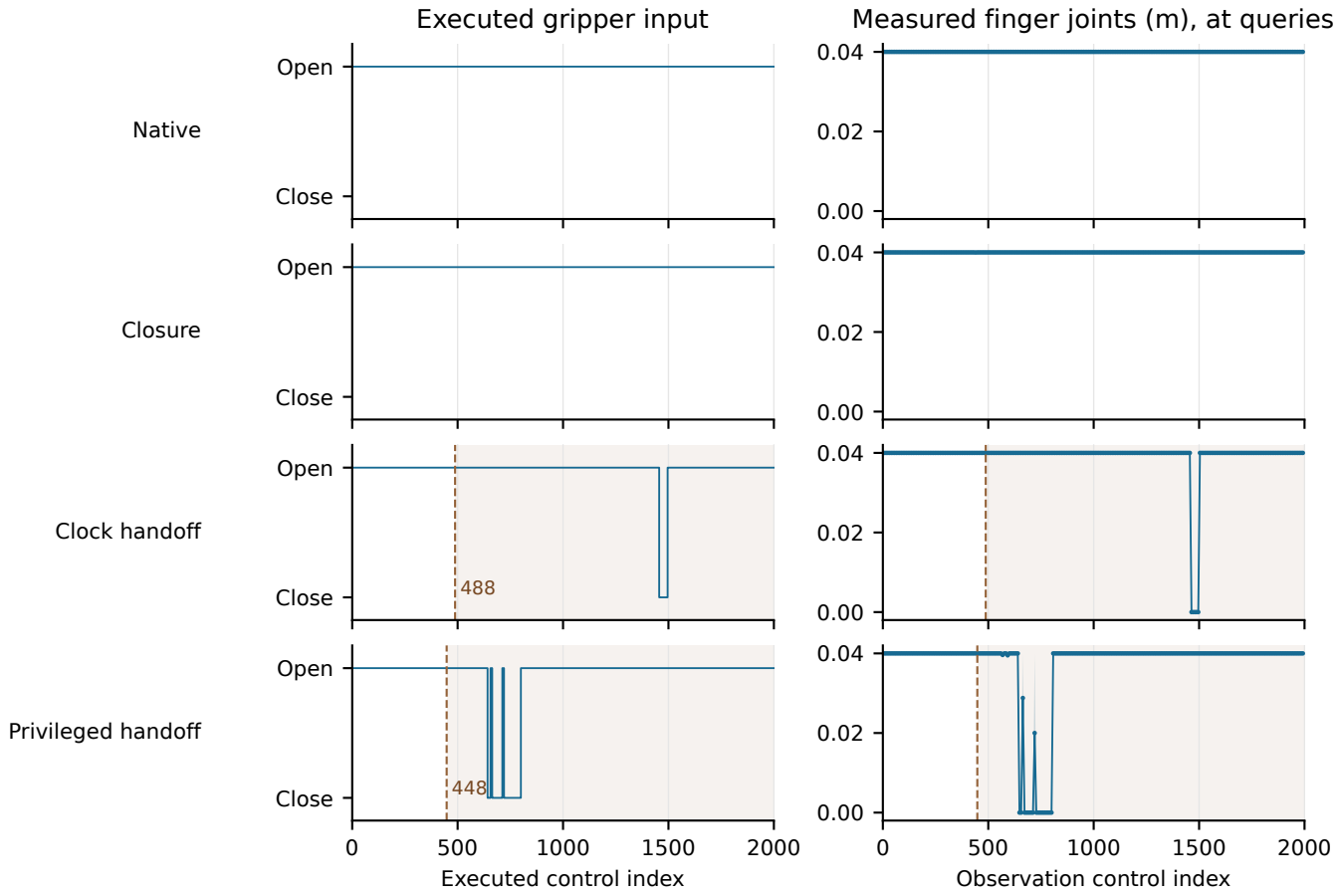


Figure 3: Actual gripper inputs and measured query-time finger joints in the four completed engineering trials. Dashed lines mark actual switches; tan regions show native continuation. At-switch positions precede the first native action. Points are two-joint means and blue bands span the two joints, without implying uncertainty, fingertip aperture or a body-link grasp. A closing input does not prove holding. Every plotted trace is bound to the completed physical audit.

tasks 0017/0013 supply 249 calibration and 83 diagnostic transitions. Splits are by whole exposed task, without checking outcomes. Their 18,666 original acquisition controls are reused, without being charged again. Risk fitting excludes already-doomed prefixes.

Actual-feature readout MAE is .191 for closure, .209 for holding, .662 for uprightness and .210 for support, above the .10 gate. Fitting has positive closure/holding in only one episode each, no in-goal positive episode or safe completion; its observed closure follows a violation. Calibration has one episode against the required minimum 20. Conservative abstention rejects every prefix; its 100%

Table 8: CPU prediction audit on one diagnostic episode. RMSE rows are correlated within that episode. The frozen model/persistence ratio gate is at most .90 in every block; all three fail.

Feature block	Model	Persistence	Ratio
Robot/history	.16805	.08516	1.973
Overview	.04381	.02848	1.538
Wrist	.06380	.05481	1.164

negative-prefix alarm rate is not a qualified classifier's false-alarm rate. Counterfactual-action coverage remains unsupported. The audit costs 27,412 training-row uses and 5,395 model-head row queries, with no new physical controls, VLA calls or encoder images. It is not refitted on the four new screen trials. The failed model remains excluded from physical action selection.

8 LIMITATIONS AND REMAINING EVIDENCE

This interface can consume several family checkpoints, but the pilot fits only within Jar. A random four-direction basis may miss useful behaviour. A bounded noise offset need not imply safe contact dynamics. The 176-parameter domain is large relative to two-case candidate feedback. Noisy selection, adverse behaviour and identity finalists are valid outcomes. If a constant explains an improvement, conditioning is unestablished; if random search matches EA, evolutionary advantage is unestablished. An EA lead over uniform random search combines parental reuse with a local Gaussian proposal distribution. Attributing it to evolutionary reuse additionally needs a fixed-parent Gaussian-search ablation and a matched local-search comparison under separate, prospectively funded evaluations.

A training-only retrospective audit identifies a concrete fitness limitation. Safe unfinished inactivity and supported closure both score zero. In generation two, EA's identifier tie-break retains a controller with no ordered progress; an already evaluated tied child closes safely on one case. A progress secondary key would change the preferred factual panel in two of five observed cohorts, including one random-search cohort. This does not reconstruct a changed search history or establish completion. A revised fitness and control representation need a new protocol and matched competitors before further physical search.

The local camera profile, selected cases and stricter requested sequence also limit comparison to published benchmark rates. Our declared profile is 640×640 DLAA; the published protocol describes 256×256 observations [5], while the pinned release evaluator defaults to 640. A source-matched CPU check on four saved initial inputs confirms the expected view/state mapping and 224×224 images after model transforms. It does not inspect the loaded live transform, normalization, tokenization or model weights, or establish pixel/training-domain equivalence. Published rates are therefore not a matched local reproduction. This changes no completed comparison and does not diagnose a wrong model-input shape or general checkpoint incapability.

An episodic parameter-REINFORCE interface over the same bounded matrix is implemented and CPU-checked, including interrupted-batch recovery. It has zero physical evaluations and is outside the frozen minimum allocation. It does not replace a competent state-conditioned DSRL-style latent SAC baseline. That baseline, matched tuning and independent optimizer replication are required before claiming practical evolutionary superiority. A direct-action residual baseline is additionally needed to attribute any advantage to the latent representation.

The cross-family target requires freezing one fitted controller and checking new tasks and an adaptation-held-out family with its own frozen base policy. Response-informed bases, persistence, sampling-distribution control and internal activation gates remain prospective, with no physical result here. A proposed 156-episode adaptation/transfer bridge is separately budgeted and unexecuted. Neither extensions nor tuning may consume the final checking allowance.

9 REPRODUCIBILITY AND PRESERVATION

The launch revision is `4783369c5b59`. Upstream pins are ManiGuard `6bf4c61a1f16`, BEHAVIOR `88454bd04f75` and openpi `215abfb217db` [6]. Separate uv-managed environments serve simulation (Python 3.10, Torch 2.6/CUDA 12.4), policy (Python 3.11, JAX .5.3) and CPU tooling. Comparative receipts retain actual GPU UUIDs, source hashes, causal features, decoded chunks, executed inputs and raw/requested outcome evidence. An independent reader verifies physical records before assigning fitness. Two early Jar development trials lack per-process GPU UUID receipts.

Code, compact evidence and frozen protocols are preserved in https://github.com/Sachin2911/Safety-Dial. The protocol is `configs/maniguard/latent-evolution/capacity-20261020-v2.json`; its study README documents reconstruction. Separate private Hugging Face model/dataset repositories store immutable stage archives with remote revision, privacy, size and SHA-256 verification. Fresh private-archive restoration re-verifies all 60 original episodes, including 112,095 exact native monitor entries and 13 authoritative source paths. The four sealed finalist matrices, all 32 original identity pairs and eight supplemental pairs also replay without new physics or model calls. Independent review reproduces all outcomes and joins all 14,019 physical client/server receipts exactly. Its decoder-only accounting has a separate scope. The four continuation starts have separate immutable archives, captured service tails and full restored-physical replay. Raw runs, licensed assets, credentials and base-policy weights are excluded from public Git.

10 CONCLUSION

Native-noise steering is verified in two family policies, but the completed Jar pilot establishes no evolutionary safe-completion advantage. Forty training trials remain requested-unfinished; four nonviolating lid closures never extend to holding. Every condition has zero safe requested completions in four protected checking cases; raw goals do not satisfy the safe requested sequence. Random retains the fixed controller's weights. A separate exposed-seed four-start screen repeats safe closure and exercises the privileged handoff, but neither handoff holds the target or completes; both violate. No next stage qualifies. The failed model remains excluded. These findings concern this representation

and budget, not all noise-space adaptation. Competent matched learning, explained completion gains and replication remain required.

ACKNOWLEDGMENTS

This BSc Honours research at the University of the Witwatersrand is supervised by Geraud Nangue Tasse. AI coding assistance supported implementation, diagnosis and drafting; recorded empirical evidence and its stated limits govern the claims.

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